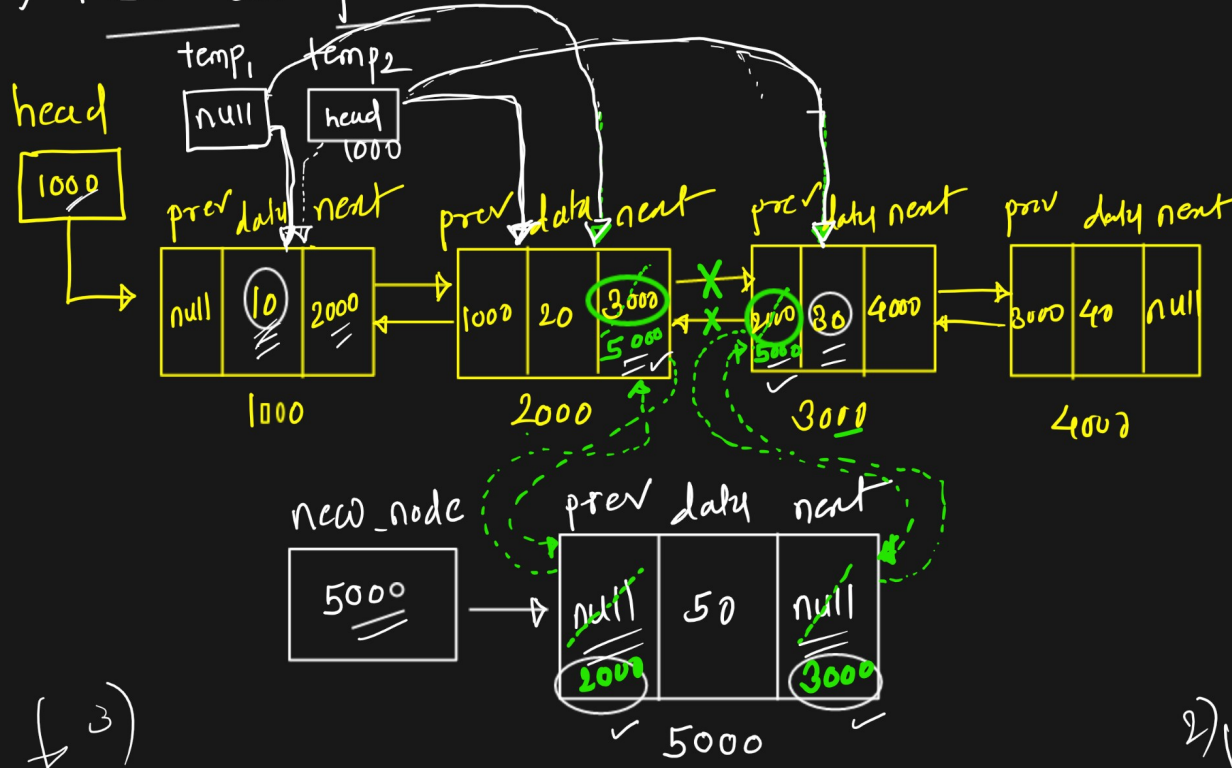


Day - 22 - Double Linked List & Circular Linked List

c) Insert at posⁿ :-



```
void insert_At_posn(int posn)
{
    30
    50
    Node new_node = new Node(50);
```

```
    if(head == null)
    {
        S.o.p("linked list empty");
        return;
    }
```

```
    Node temp1 = null;
    Node temp2 = head;
```

```
    while(temp2.data != posn && temp2 != null)
    {
        temp1 = temp2;
        temp2 = temp2.next;
```

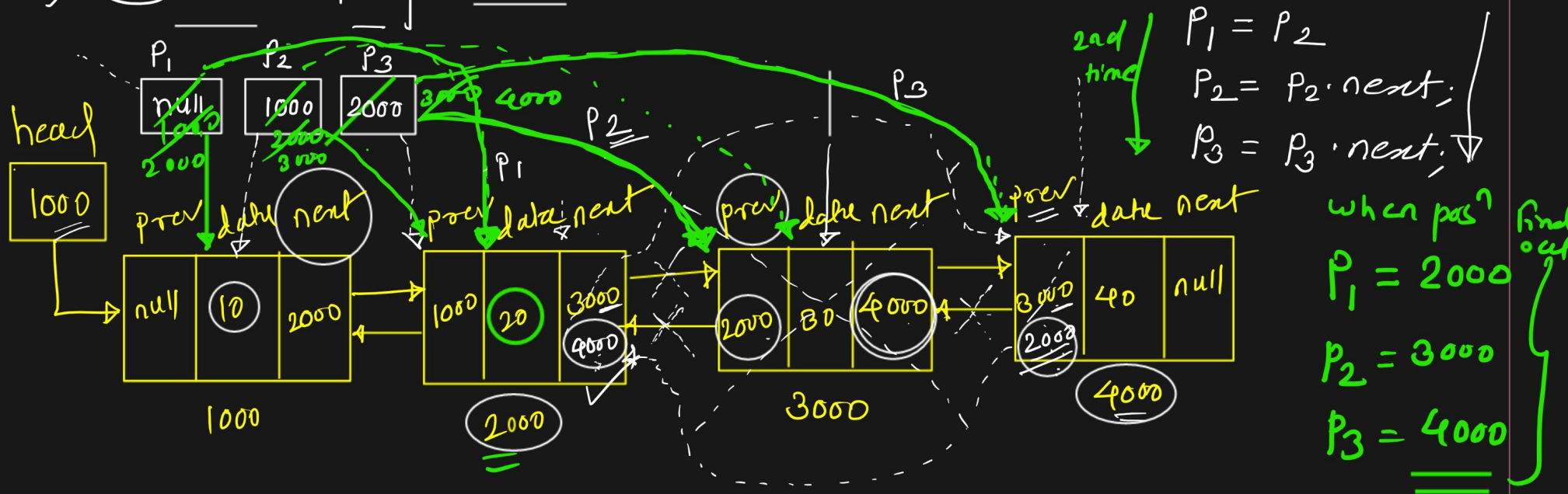
2) continue

```
    if(temp2 == null)
    {
        S.o.p("posn value doesn't found");
        return;
    }
```

```
    new_node.prev = temp2.prev;
    new_node.next = temp1.next;
```

```
    temp1.next = new_node;
    temp2.prev = new_node;
```

D) Delete Any Node :-



```

void delete_any_node (int value)
{

```

```

    if (head == null)
    {
        s.o.p ("Empty");
    }
    return;

```

else

```

{
    Node p1 = null; // null
    Node p2 = head; // 1000

```

Node $p_3 = \text{head.next}; // 2000$

```

while ( p2.data != value && p2.next != null )
{

```

$p_1 = p_2;$

$p_2 = p_2.next;$

$p_3 = p_3.next;$

}

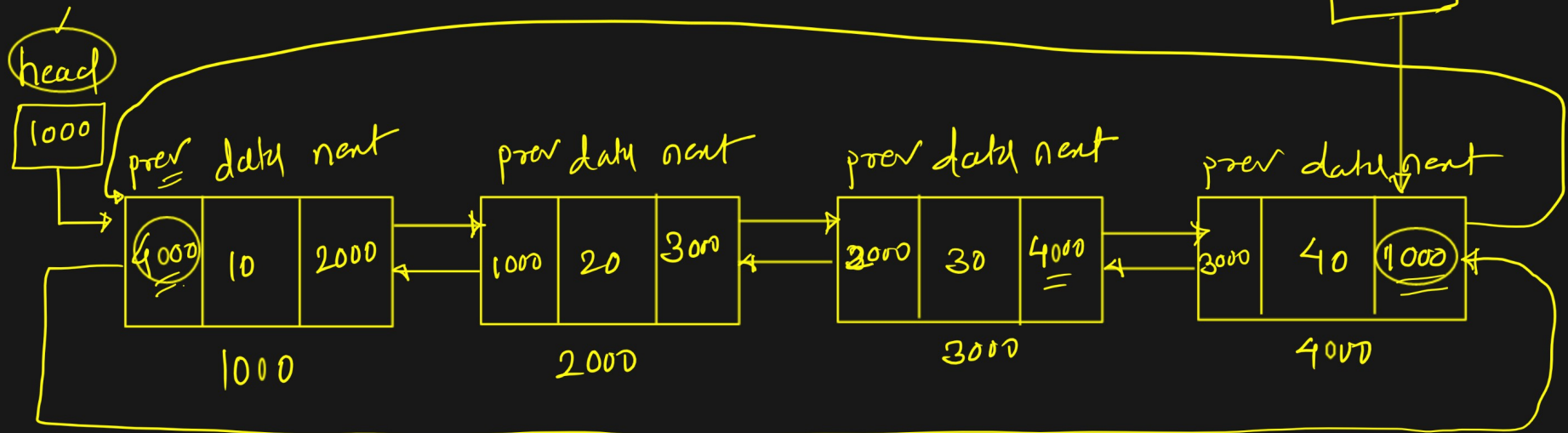
$p_1.next = p_2.next;$

3000

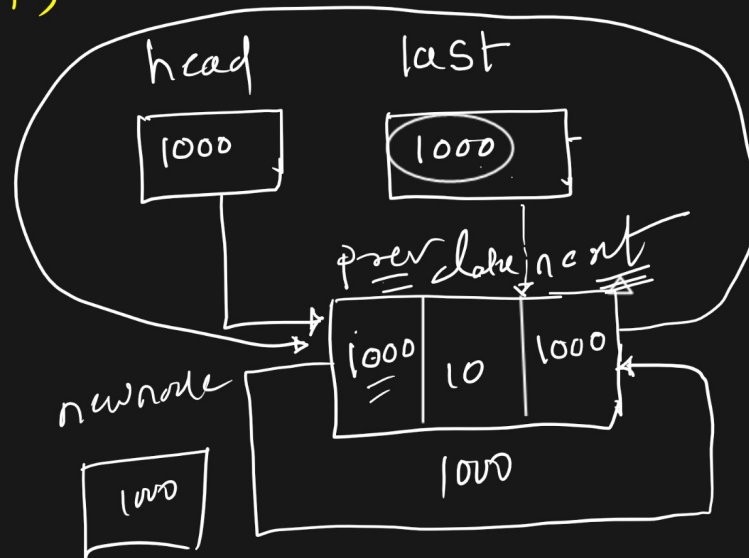
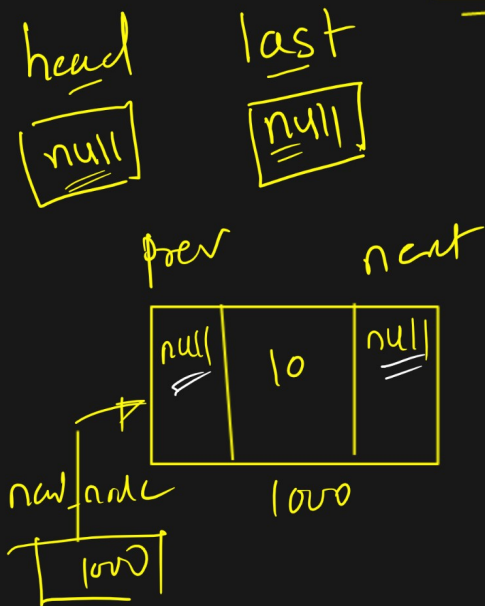
LL empty

$$p_3.\underline{\underline{prev}} = p_2.\underline{\underline{prev}};$$

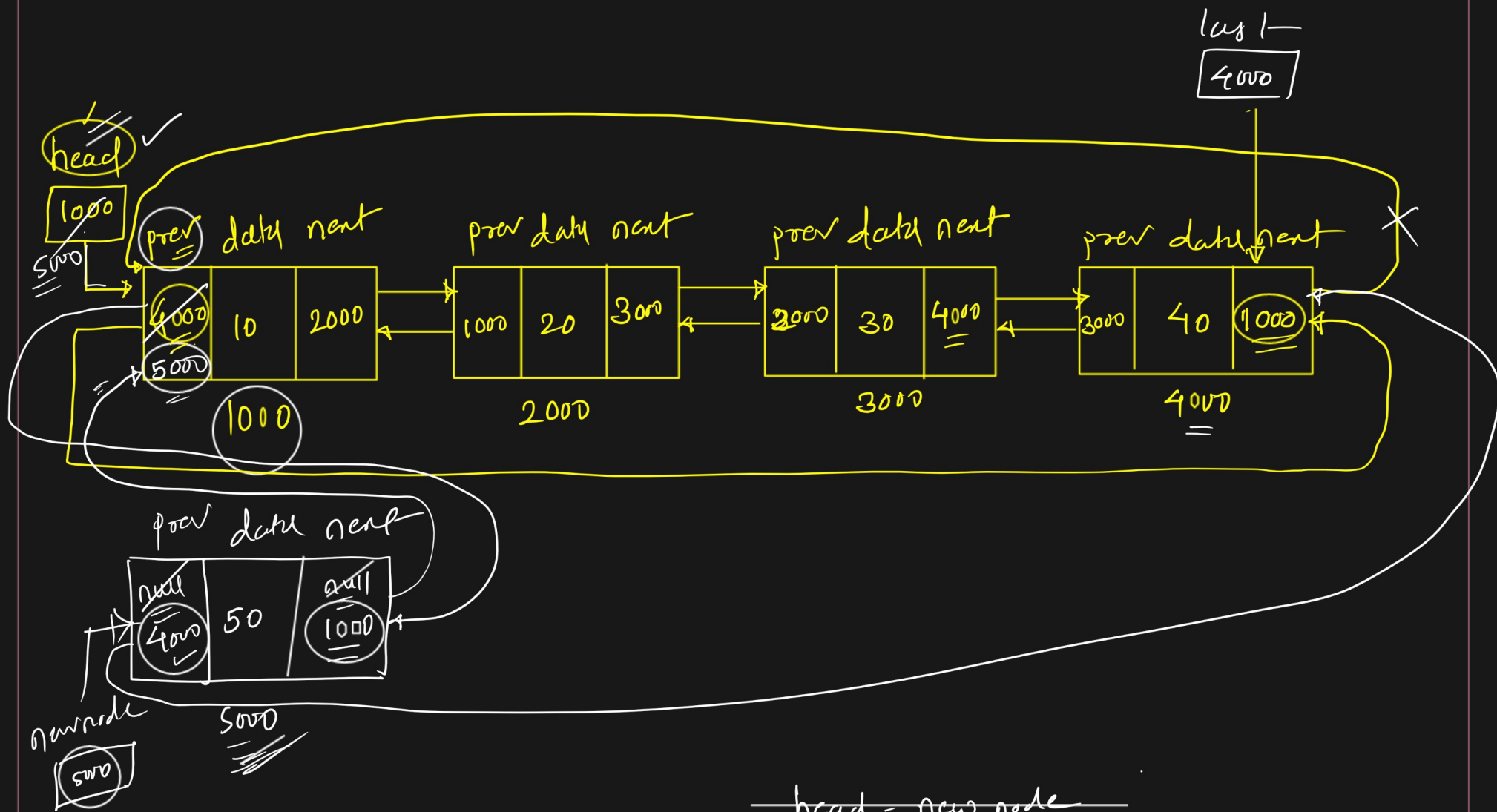
* Circular Linked List:-



1) IF linked list is empty



$head = new_node$
 $last = new_node$
 $head.\underline{\underline{prev}} = \underline{\underline{new_node}}$
 $last.\underline{\underline{next}} = head$



CLL

Insert At First $\Rightarrow$ head $\rightarrow$ last.next

Insert At last $\Rightarrow$ last $\rightarrow$ head.prev

Delete At first $\Rightarrow$ head $\rightarrow$ last.next

Delete At last $\Rightarrow$ last $\rightarrow$ head.prev

